

K1 MUSE Pi Pro

RISC-V Architecture For Intelligent Future



Compact and efficient, powering the future of AI and IoT

K1 MUSE Pi Pro is a single-board computer that integrates a high-performance RISC-V 8-core processor, storage, I/O, and expansion capabilities into a credit card-sized 1.8-inch board. With UEFI support, open-source resources, and strong compatibility, it is ideal for AI applications, robotics, education, and IoT scenarios requiring efficient power and compact form.



Compact Design

Credit card-sized 1.8-inch single-board computer



UEFI Support

Flexible boot media selection and system configuration



Fully Open Source

Open board-level, mechanical, and software design resources with application guidance



GRUB Boot Support

Multi-OS installation with user-selectable boot options



Domestic Innovation

Locally developed solution ensuring supply continuity



Standard I/O Expansion

miniPCIe and M.2 interfaces compatible with mainstream components



High Reliability

Structural protection, ESD resilience, and EMC compliance



Specifications

Processor	SpacemiT M1 with 2.0 TOPS AI power
Display	HDMI 1.4 (up to 1080P @ 60 Hz) 2-lane MIPI DSI FPC (up to 1080P @ 60 Hz)
Memory	LPDDR4X @ 2400 MT/s, 8 GB / 16 GB options
Local Storage	eMMC 5.1, 64 GB / 128 GB options
Storage Expansion	M.2 M-Key connector, 2230-size NVMe SSD MicroSD slot, supports UHS-II
Wireless	Onboard Wi-Fi 6 and Bluetooth 5.2, supports dual wireless via M.2 / mPCIe
Network Port	1 × Ethernet port, RJ45 connector, 1000 M / 100 M auto-sensing
Audio	3.5 mm audio jack
USB	4 × USB 3.0 Type-A (host), 1 × USB 2.0 Type-C (device)
Debug	UART TTL debug port + 3 buttons: reset / power / flash
I/O Expansion	Standard M.2 M-Key slot (2230, supports SSD, PCIe-to-SATA, PCIe communication boards, etc.) Full-size miniPCIe slot (supports 4G / 5G, PCIe wireless / network modules) 40-pin standard GPIO interface
MIPI Interfaces	1 × 2-lane MIPI DSI FPC 15-pin interface 1 × 4-lane MIPI CSI FPC 22-pin interface, 1 × 2-lane MIPI CSI FPC 15-pin interface
Form Factor	Single-board computer, FEMTO-ITX dimensions, 85 × 56 mm (credit card size)
OS	Bianbu , Ubuntu, OpenKylin, Deepin, Fedora and other OSs
Power	USB-PD: 5 V / 9 V / 12 V @ 3 A
Reliability	ESD: ±4 kV contact / ±8 kV air, CE / FCC compliant Available in consumer-grade (-20 °C to 70 °C) or industrial-grade (-40 °C to 85 °C) versions
Structure	Optional passive or active cooling heat sinks Optional metal chassis or acrylic case Optional touch screen or industrial terminal configurations